

# Sidhant Maloo

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**Software and hardware builder** with completed projects across machine learning, simulation, PCB design, and robotics. Teaches Java and Unity as a Code Ninjas Sensei. Seeking internship opportunities in software or hardware engineering.

## TECHNICAL SKILLS

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|-------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Software</b>   | Python, Java, JavaScript, HTML/CSS, Qiskit; machine learning (SVR, Random Forest, backprop neural networks) |
| <b>Algorithms</b> | Inverse kinematics, pathfinding, numerical simulation, machine learning model comparison                    |
| <b>Hardware</b>   | PCB design (KiCad), circuit simulation (Multisim), soldering, Raspberry Pi, VEX V5, mechanical design       |
| <b>Sensors</b>    | GPS, vision, bumper, color, and TTP223 capacitive touch sensors                                             |

## PROJECTS

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**Humanoid Robot ("Bob")** *Python* May 2026 – Present

- Designing an AI-controlled upper torso: two 3-DOF arms and a head with a camera, microphone, and speaker that acts on spoken requests.
- Software is largely complete; integrating the physical body next.

**Interactive Simulations** *six completed* 2026

- Built six interactive simulations spanning quantum computing, black hole dynamics, robot-arm kinematics, neural networks, pathfinding, and circuit physics, with live visualizations and error tracking.
- Published with code on his website and GitHub, so anyone can run and inspect the work.

**Machine Learning Stock Model & Research Paper** *Python* 2026

- Built a model that scores a stock's potential, then tested how SVR and Random Forest hold up as noise and volatility increase.
- Wrote up the findings and posted them publicly on Zenodo; code is on GitHub.

**Hardware & Electronics** 2020 – 2026

- Designing a custom PCB in KiCad; built a Raspberry Pi cyberdeck, a soldering-station fume extractor, and a TTP223 touch-controlled light.
  - Diagnosed and repaired a Sky Viper drone by tracing faults through an unfamiliar system without instructions or a blueprint.
- Also:** additional software projects (a Java game, teaching demos, system tools) on GitHub.

## EXPERIENCE

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**Code Ninjas, Algonquin, IL** — Sensei, Upper Belt Lead Apr 2026 – Present

- Supervise and tutor students across a wide age range in Unity, Java, and core software skills, turning technical concepts into clear lessons.

**VEX V5 Robotics State Team** — Builder, Designer & Programmer Aug 2024 – May 2026

- Programmed autonomous and driver-controlled behavior from sensor data; contributed to mechanical design and iterative prototyping.
- Helped earn the **Think Award** for documented engineering design. When funding and coaching fell through for 2026-27, sent dozens of emails and met with the school board to keep the club alive.

## LEADERSHIP

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- **Captain, State Math Team:** lead practices for six students; 2× regional champion and state competitor.
- **Treasurer, North Central District Student Council:** organize large groups of students from schools across Illinois.
- **Student Rep., Principal Student Leadership Council:** represent students on policy affecting hundreds of peers.

## EDUCATION & HONORS

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**Huntley High School** — Junior, GPA **3.97** unweighted / **4.62** weighted Expected May 2028

- AP Computer Science A (5), Computer Science Principles (5), Precalculus (5), Physics 1 (4); PLTW Digital Electronics; AP Calculus BC; MIT OpenCourseWare 18.01 (completed).
- UN Ambassador Award (2026); Illinois Principals Association Student of the Year (2024); Illinois House of Representatives Honors (2025).